Teamwork guide: dataset preparation, preprocessing and EDA

DSC 207 | Revised milestone scope | Owners unassigned

This guide matches the four new notebooks in notebooks/milestone_preprocessing_roles/. It is a plan for work to be completed, not evidence that the team has completed these tasks. The earlier notebooks, reports and teamwork guides are preserved.

What this milestone covers

The project question remains whether four-parameter inference generalizes from O3a H1 noise to O4a H1 noise. This checkpoint concentrates on preparing and understanding the dataset: source provenance, physical labels, preprocessing, actual split counts, EDA, plots, and limitations. Model training, model performance tables and calibration conclusions are outside this notebook set's required work.

The methodology section must still describe three algorithms because the rubric asks for a baseline and two improvements: linear regression, its small regularized extension Ridge, and SNPE with a masked autoregressive flow. Describe them as the next common comparison; do not invent new training results. Feature importance is explicitly requested by the rubric. Input-target association is useful EDA but does not fully substitute for model-based importance; Role 4 must state this gap honestly if importance is deferred.

Existing evidence versus your work: The repository already contains saved noise, injections and processed data. The revised PDF uses those artifacts, not results from these new notebooks. Your task is a separately identified manual pilot. Do not claim your pilot has the old dataset's 800-per-run counts until you actually generate and check them. The earlier split was a 750-row O3a development pool, 50 O3a held-out examples and 800 O4a evaluation examples. A 600/150 subdivision of the development pool is a proposal, not a completed split.

Implementation rule: Write your own code in the empty cells. You may call standard scientific libraries and consult their documentation, but do not import ligo_pipeline, project utils, config.py, another notebook's implementation, or prepared training/model checkpoints. Trusted raw strain with source metadata is allowed. You may write your own small functions after explaining their steps. No solution code is supplied here.

Ownership: Assign one person per notebook; names are unassigned. Review each other's outputs, but let each owner make their own notebook changes. Record actual contributions and dates. Every cell has a sequential cell_number and every markdown cell displays its number; the preceding task labels the empty code cell. Renumber all cells from 1 when inserting new cells. Execution counters are not stable cell numbering.

Four roles and the evidence each owns

Role	Main responsibility	Milestone evidence
1	Raw data, provenance and QC	Source inventory, raw counts, raw-time/distribution plots
2	Simulation, labels and SNR selection	Injection/target tables, attempt ledger, population EDA
3	PSD, whitening, filtering and cropping	PSD handoff, processed inputs, quality checks and plots
4	Dataset audit, EDA and methodology plan	Reconciled counts, input-target plots, planned comparison

Agree on handoffs before expanding the pilot

Common contract and parallel work

Agree on one small experiment before scaling up: H1; O3a development and within-run evaluation; O4a cross-run evaluation; 4096 Hz; four-second final windows; a 35-350 Hz preprocessing band; target order chirp_mass, mass_ratio, chi_eff, distance. Keep any fixed-mass, distance-only exercise separate from this four-target pilot.

Use a new experiment_id and store files under artifacts/manual_preprocessing_milestone/<experiment_id>/. The role-owned subfolders are data/, simulation/, preprocessing/ and analysis/. Put experiment.json in data/ and repeat its ID and relevant settings in each output's metadata. Store raw segment duration, sample rate, waveform assumptions, parameter ranges, seeds, intended counts, actual counts and units. Do not overwrite the existing artifacts or the older role notebooks.

Role 1 supplies raw arrays, noise_id/group_id, source GPS intervals and source split assignments. Role 3 supplies noise-only PSD files early. Role 2 uses those PSDs for SNR selection, then supplies padded injected buffers, merger indices and labels. Role 3 supplies processed arrays and row metadata. Role 4 audits and combines evidence, counts and EDA; this role does not train the final models.

Source-group separation: Assign groups before simulation. Overlapping raw GPS intervals are one group even if filenames differ. Use train, validation, O3a_test and O4a_test split names; keep all O4a out of fitting and tuning. A recommended small design has at least three O3a training groups, one O3a validation group, one O3a test group and two O4a groups. Choose feasible durations and target counts together; these are planning suggestions, not completed allocations. If your pilot cannot support this design, document the limitation. Track nonoverlapping final crop windows as well as group identities.

Handoff format: Use .npy/.npz numeric arrays and .csv/.json metadata with repo-relative paths. Join per-example records by unique sample_id; join noise/PSD records by noise_id. Preserve run, split, group and units. Never assume two files have matching row order. Save rejection ledgers, shapes, dtypes, versions and seeds. Changes to schema/settings require all consumers to check compatibility.

Start in parallel: Role 1 can inspect one raw segment, Role 2 can inspect clean waveforms, Role 3 can validate sinusoids/FFT/filtering, and Role 4 can design the counting ledger, plot checklist and split audit. First integrate one real example end to end through data preparation. Then expand the pilot; no model fit is needed to check these handoffs.

Integration order

Raw segment (Role 1) -> noise-only PSD (Role 3) -> aligned injection and labels (Role 2) -> processed row (Role 3) -> dataset audit and EDA (Role 4). The early PSD delivery prevents the SNR calculation from blocking on the full preprocessing dataset.

First check one real example, then expand to the agreed pilot. Everyone can begin the independent exercises while waiting for upstream files. There is no need to fit an end-to-end model to demonstrate these data-preparation handoffs.

Role 1 - Data acquisition, provenance, and raw-data EDA

Build trustworthy raw inputs and freeze a defensible group split before anyone fits a model. You own acquisition, raw quality checks, source documentation, and raw-noise EDA, not simulation or preprocessing. Your checkpoint deliverable is data evidence, not a trained model.

Notebook: 01_raw_data_provenance_and_eda.ipynb

Inputs: Public GWOSC H1 O3a/O4a strain or trusted local raw .npy files with original metadata. A local downloaded file is a source, not evidence that its quality is already verified.

Start in parallel: Start with one public/local raw segment, a tiny hand-created metadata table, and descriptive plots. Freeze the table schema immediately. Deliver one vetted segment per available run early, then add split groups. Role 3 can start PSD work from the first accepted segment; Role 2 can simulate waveforms before all noise arrives.

Task 1: Understand strain and provenance

Background: Strain is a dimensionless time series. Sample k occurs at t0 + k/fs; N/fs is its duration, and fs/2 is the Nyquist frequency. UTC and GPS are different time standards. A long segment has millions of samples but is not millions of independent labeled examples.

Your task: Select the pilot sources, explain UTC/GPS fields, list the exact source locations, and record measured N, fs, duration, units and dtype. Check metadata against the array. Create your experiment record and explain which settings are fixed versus provisional.

Evidence: A source inventory, exact sample counts and one correct time axis.

Task 2: Acquire and audit raw backgrounds

Background: Real noise can contain gaps, invalid samples, transients and known astrophysical events. Catalog exclusions and quality flags reduce contamination but do not prove a segment is glitch-free. Missing strain is not a categorical value to encode.

Your task: Read/fetch candidate segments yourself. Check shape, finite fraction, mean/std/min/max and available data-quality information. Explain your rejection rules and event exclusion policy. Retain an accepted/rejected ledger. Reject invalid data rather than silently filling gaps; if you choose interpolation, justify why it is scientifically suitable.

Evidence: QC table with actual before/after counts, reasons and a clear limitation when quality flags are unavailable.

Task 3: Inspect raw distributions and time behavior

Background: A histogram reveals asymmetry or heavy tails; a full time plot may hide short transients. Blockwise mean/std reveal changing scale. Raw colored noise is not expected to have equal power at every frequency.

Your task: Plot full time series, a short zoom, a histogram and blockwise statistics with time in seconds and strain on the correct axes. Compare runs when both are available. Interpret at least two patterns; do not claim stationarity from one plot alone.

Evidence: At least three properly labeled plots and short interpretations.

Role 1 - Tasks and handoffs, continued

Task 4: Freeze source groups and the split

Background: Injections sharing the same background can share noise patterns. Randomly splitting their rows can exaggerate generalization. Group by original nonoverlapping GPS intervals before simulation; reserve validation for selection and tests for evaluation.

Your task: Assign each accepted source group one split. Check GPS overlap, prohibit group reuse across O3a splits, and keep O4a test-only. Export the split manifest. Record intended example counts; identify that actual counts will be measured after injection and preprocessing.

Evidence: A group-disjoint split audit and source-duration/sample-count table per run and split.

Task 5: Export one early handoff, then the pilot

Background: A downstream handoff is a stable data contract. Files need explicit provenance so another person can inspect them without knowing your variable names.

Your task: Export raw .npy files and the manifest. Ask Roles 2 and 3 to verify one row and array. Correct your own notebook from their feedback; then extend the pilot. Record source hashes if practical and versions/seeds.

Evidence: Role 2 and Role 3 can load one segment and verify fs, units, noise_id and split without importing your code.

Task 6: Prepare your milestone evidence

Background: Data claims need separate counts for raw time samples, quality-screened segments and labeled examples. You can report raw counts now and receive example counts from the other roles.

Your task: Write 150 words on source, size, QC, temporal patterns and data challenges. Include a plot caption and source link. Update your contribution log and notebook numbering. State what has actually run and what remains planned.

Evidence: A raw-data evidence bundle and a contribution entry for this notebook alone.

Your exported files and acceptance contract

Outputs: data/experiment.json; data/noise_manifest.csv; data/noise/<noise_id>.npy; data/split_summary.csv; labeled raw-data figures and a 150-word Data/Challenges draft. Paths are under artifacts/manual_preprocessing_milestone/<experiment_id>/.

File contract: noise_manifest.csv columns: noise_id, group_id, run, detector, gps_start, gps_end, sample_rate_hz, n_samples, strain_path, split, source_url_or_key, quality_status, quality_reason. gps_end is exclusive. noise_id is unique. Record all rejected candidates separately with reasons; do not delete them from the acquisition accounting.

Use the experiment-specific root artifacts/manual_preprocessing_milestone/<experiment_id>/ for all paths. Confirm sample IDs, source groups and retained counts with the next owner.

Role 2 - Waveform simulation, injection, and label EDA

Create labeled observations from physical parameters and real noise. You own parameter draws, waveform generation, placement, target derivation, SNR accounting and accepted-population EDA. Your checkpoint deliverable is data evidence, not a trained model.

Notebook: 02_simulations_labels_and_selection.ipynb

Inputs: Role 1 raw noise and frozen source splits; Role 3 noise-only PSD files. Start independently with waveforms and a zero-noise or simple-noise practice buffer, clearly labeled as an exercise.

Start in parallel: Draw parameter tables and compare clean waveforms immediately. Practice alignment on a small known array. Role 3 supplies an early PSD artifact; use it for a small real-noise pilot before increasing the number of attempts. Do not estimate production PSDs here.

Task 1: Define a physical pilot and draw parameters

Background: A prior is the distribution of physical parameters before seeing data. Sorting component masses enforces m1 >= m2. A distance draw proportional to d squared is Euclidean volume weighting; it is not an exact cosmological comoving-volume calculation. Derived targets are usually not uniform even if components are.

Your task: Choose and record mass, aligned-spin and distance ranges. Match the existing ranges if feasible: 5-100 solar masses, spins 0-0.8, distance 100-2000 Mpc. Write the draw yourself, fix a seed, and create at least 100 practice rows. Explain any narrower pilot range. Discuss with Role 4 how the simulation distribution relates to an inference prior.

Evidence: Prior table, parameter bounds, units and empirical histograms; no claim of uniform derived targets.

Task 2: Explore waveform physics before injection

Background: IMRPhenomD models aligned-spin inspiral, merger and ringdown. h_plus and h_cross are polarizations, not automatically the detector response. At fixed remaining parameters, amplitude scales approximately as 1/distance. For a learning pilot, using h_plus directly is a simplified convention that must be documented.

Your task: Call the waveform library directly. Inspect time spacing, duration and peak location. Compare distances 100, 500 and 1000 Mpc with identical masses/spins and a common vertical scale. Vary mass separately. Agree on a fixed inclination/polarization convention; do not claim a full detector projection if you use plus-only.

Evidence: Clean-waveform comparison figure with correct time/strain axes and an explanation of amplitude versus timing changes.

Task 3: Derive the four targets

Background: Chirp mass Mc = (m1*m2)^(3/5)/(m1+m2)^(1/5), mass ratio q = m2/m1, and effective spin chi_eff = (m1*chi1+m2*chi2)/(m1+m2). Distance is kept in Mpc. These labels come from the simulator, not measured noise.

Your task: Compute the target table yourself. Audit mass ordering and formula results on a few simple manually checked cases. Preserve sample IDs and the shared target order. Explain why pure noise does not have a distance/mass label of zero.

Evidence: Finite target table with units, range checks and manually verified examples.

Role 2 - Tasks and handoffs, continued

Task 4: Align and inject without changing source noise

Background: Injection means y = n + h on a common time grid. Align the waveform peak, not its last sample, to the selected merger position. Ringdown continues after the peak. A sufficiently padded buffer lets preprocessing remove edge artifacts before the final crop.

Your task: On a tiny practice buffer, demonstrate the placement indices. Then create a new copy of each real buffer, add the waveform, and retain the original noise unchanged. Check both waveform boundaries, crop clearance and nonoverlapping final windows. Preserve local and source indices. Inherit the source split and group.

Evidence: Aligned noise/waveform/injection plot, placement metadata and boundary checks.

Task 5: Compute SNR and track selection

Background: Optimal SNR squared is 4 times the positive-frequency integral of |h_tilde(f)| squared / PSD(f). FFT normalization, sample spacing, frequency spacing and PSD units must agree. A waveform norm is not the same as a matched-filter search over noisy data. An SNR gate changes the accepted population.

Your task: Use Role 3 PSDs to calculate optimal SNR. Implement a frequency-sum check and compare with a library calculation on one example. Specify low/high cutoffs separately from the preprocessing band; record any departure from the existing 20 Hz lower SNR cutoff. Apply an agreed threshold, such as 8, with a finite attempt budget. Count every attempted, accepted, rejected and failed draw by run and split.

Evidence: Validated SNR convention and complete selection ledger; insufficient accepted examples are reported rather than hidden.

Task 6: Export labels and analyze selection

Background: Accepted injections are a selected population. Different PSDs can produce different rejection rates even with identical draw rules. Target correlations may arise from shared component parameters, not learned model importance.

Your task: Export buffers/manifests/targets. Plot accepted target and SNR distributions by run; compare attempted versus accepted parameter draws where possible. Make a target correlation heatmap and a distance-SNR scatter plot. Interpret selection effects and confirm that Role 3 can reconstruct one buffer/time position. Prepare 150 words and a contribution log.

Evidence: Detailed population EDA, actual counts and an accepted-ID handoff; do not assert noise-induced drift from these plots alone.

Your exported files and acceptance contract

Outputs: simulation/injection_manifest.csv; simulation/attempts.csv; simulation/buffers/<sample_id>.npy; simulation/targets.csv; population/SNR figures and simulation notes. Keep full padded buffers for Role 3, not final cropped windows.

File contract: injection_manifest.csv: sample_id, noise_id, group_id, run, split, buffer_path, buffer_start_in_noise_samples, buffer_n_samples, merger_index_in_buffer, waveform_start_in_buffer, waveform_n_samples, snr, status. targets.csv: sample_id, mass_1, mass_2, chi_1, chi_2, distance, chirp_mass, mass_ratio, chi_eff. attempts.csv also records rejected/failed attempts and reasons. The inherited split cannot be changed.

Use the experiment-specific root artifacts/manual_preprocessing_milestone/<experiment_id>/ for all paths. Confirm sample IDs, source groups and retained counts with the next owner.

Role 3 - Preprocessing and signal validation

Own the complete preprocessing chain: PSD, whitening, bandpass, crop, and numerical/physical checks. This is the dedicated preprocessing role. You do not acquire production data, create labels or train models. Your checkpoint deliverable is data evidence, not a trained model.

Notebook: 03_preprocessing_and_quality_checks.ipynb

Inputs: Role 1 raw noise and source metadata; Role 2 padded injected buffers with merger indices and target IDs. Until those arrive, use simple sinusoids/noise and one real raw segment to learn the transformations.

Start in parallel: You can study FFTs, validate frequency axes, design a filter and implement crop boundary checks immediately. Export a noise PSD as soon as Role 1 supplies one segment. Finish real whitening/cropping once Role 2 supplies one padded injection. The full dataset is not needed to prove your methods.

Task 1: Build the time and frequency foundations

Background: FFT bins are fk = k*fs/N, the Nyquist frequency is fs/2, and frequency resolution is fs/N. A raw FFT magnitude is not a PSD. For real inputs, a one-sided transform avoids duplicate negative-frequency bins. Windowing changes spectral leakage.

Your task: Construct a known sinusoid with a duration and sample rate you choose. Compute its frequency axis and show where its peak should be. Compare windowed/unwindowed spectra, then inspect one real segment. Explain bin spacing, Nyquist and the 35-350 Hz analysis band.

Evidence: Time/spectrum plots and a measured sinusoid peak error; axes include seconds, Hz and units.

Task 2: Estimate a noise-only PSD and export it early

Background: Welch PSD averages overlapping windowed periodograms and has units strain squared per Hz. ASD is its square root. More averaging reduces variability at the expense of frequency resolution. Estimate background PSD from the source noise, not from a buffer containing the injected signal.

Your task: Implement and inspect a small Welch-style calculation, then compare with a trusted library result. Use four-second subwindows when the raw segment is long enough; record overlap, window and detrending. Check increasing frequency, nonnegative PSD and units. Explain a positive floor for unusable/near-zero bins. Export the first PSD and its settings to Role 2.

Evidence: A trustworthy PSD artifact, PSD/ASD plot, normalization check and documented floor/invalid-bin policy.

Task 3: Whiten without erasing physical signal

Background: Whitening divides frequency components by the square root of a matching PSD, then transforms back. Constants depend on one-/two-sided PSD and FFT conventions. It is not feature-wise StandardScaler normalization, and an arbitrary unit-variance result alone does not prove correctness.

Your task: Choose and explain a normalization convention. Interpolate the noise PSD to the buffer FFT grid and apply whitening yourself. Compare spectra before and after on a noise-only interval. Check finite values and reconstruction behavior. Preserve the PSD scale convention for all runs; explain any floor or zero-bin handling.

Evidence: A derivation in your words, before/after PSD and time plots, and empirical checks of frequency flattening.

Role 3 - Tasks and handoffs, continued

Task 4: Apply and validate the bandpass

Background: A bandpass removes frequencies outside the analysis band. Forward/backward filtering reduces phase delay but has boundary effects; the effective response is applied twice. Second-order sections can improve numerical stability. Zero-phase filtering is an offline operation.

Your task: Design a fourth-order Butterworth 35-350 Hz bandpass for fs=4096 Hz. Plot its response. Test a below-band, in-band and above-band sinusoid and measure attenuation. Inspect both edges and decide how much padding is needed. Record design order and forward/backward behavior.

Evidence: Response plot, three-sinusoid check, edge inspection and exact settings.

Task 5: Crop only after processing padded buffers

Background: The project convention is two seconds before plus two seconds after merger: four seconds total, or 16384 samples at 4096 Hz. Cropping first can create artificial filter/FFT boundaries near the signal. Local buffer indices differ from GPS/source indices.

Your task: Use Role 2 local merger indices to crop after whitening and filtering. Verify the peak alignment on one known injection. Check exact length, finiteness and enough padding for your documented edge policy. Reject invalid bounds rather than silently truncate. Explain what happens for a near-edge candidate.

Evidence: One final input plot with merger marker, sample-count proof and a rejection ledger.

Task 6: Assemble data without changing labels or splits

Background: A target row must stay paired with the same observation through all filtering and reordering. Training-only ML scaling belongs to the model workflow; per-segment background PSD estimation is signal processing and must use a consistent inference-time policy.

Your task: Join buffers and labels by sample_id. Preserve group and split. Export X, theta and ID/split arrays plus a row manifest. Audit every join and give actual per-split counts. Do not add feature scaling fitted across all rows or normalize each waveform amplitude casually; amplitude helps infer distance.

Evidence: Role 4 can load fixed-shape inputs and labels and verify ID order without using your functions.

Task 7: Explain the evidence for the milestone

Background: A processing description must distinguish invalid samples, valid loud observations, normalization, feature selection and categorical encoding. It should explain how the transformations affect the scientific signal.

Your task: Build one multipanel raw/whitened/filtered/cropped figure, with units and matching time axes where possible. Write 150 words covering missing values, outliers, whitening, filtering, cropping and numerical issues. State no categorical encoding if inputs are numeric, and no learned feature selection if none was performed. Keep your contribution log.

Evidence: A self-contained preprocessing section, labeled figure, settings file and explicit completed/planned status.

Your exported files and acceptance contract

Outputs: preprocessing/psd/<noise_id>.npz; preprocessing/processed_manifest.csv; preprocessing/processed.npz; preprocessing/rejections.csv; before/after plots and processing_settings.json. Deliver PSD files early so Role 2 can calculate SNR.

File contract: Each PSD .npz contains frequency_hz and psd_strain2_per_hz and is identified by noise_id; its metadata records source and Welch settings. processed.npz contains X with shape (N,16384), theta with shape (N,4), sample_id and split arrays. processed_manifest.csv adds row_index, sample_id, noise_id, group_id, run, split, n_samples, finite_status and source buffer path. processed arrays and manifests must agree exactly; use explicit numeric/string dtypes, not pickled objects.

Use the experiment-specific root artifacts/manual_preprocessing_milestone/<experiment_id>/ for all paths. Confirm sample IDs, source groups and retained counts with the next owner.

Role 4 - Dataset audit, EDA synthesis, and methodology plan

Own the final dataset audit, input-target EDA, accurate count tables and methodology plan. Assemble the evidence for the three-page milestone. This role does not require model training or posterior sampling.

Notebook: 04_dataset_audit_eda_and_methodology_plan.ipynb

Inputs: Role 1 source/split manifests, Role 2 injection/target/attempt tables, Role 3 processed arrays and rejection ledger, and each owner's actual findings and figures. Use hand-created ID tables for early practice, clearly labeled as exercises.

Start in parallel: Before the data arrives, design the audit columns, draw the source-to-example lineage on paper, specify figure labels and draft the methodology/metrics plan. Test ID joins and split-overlap detection on a tiny self-created table. Replace exercises with measured pilot evidence later.

Task 1: Trace source data to final examples

Background: A raw time sample, source segment, attempted injection and processed ML row are different counting units. If an injection is rejected or preprocessing fails, it cannot silently reappear in final counts. A one-to-one ID join protects observation-label pairing.

Your task: Design a lineage table first. Join the real manifests by their IDs when available. Check unique IDs, missing labels, duplicate rows, array/manifest agreement and rejection reasons. Reconcile attempted = accepted + SNR-rejected + waveform-failed plus any separately documented failure category. Reconcile accepted inputs with retained and rejected processed rows. Do not fill unknown counts with zero.

Evidence: A reconciled audit ledger showing where every example went, with unresolved items explicitly listed.

Task 2: Verify exact split sizes and independence

Background: Train and validation serve future fitting/selection; O3a_test is a within-run holdout and O4a_test a cross-run holdout. Disjoint example IDs alone do not prevent shared-background leakage. The old 750/50/800 dataset and your new pilot are separate experiments.

Your task: Count actual retained examples and source groups per split. Check GPS/source-group overlap and nonoverlap of final crop windows. Verify O4a is never assigned to train or validation. Report whether only an allocation exists or fitting has actually occurred; model fitting is not required here. Record the old allocations only as historical context. Do not present 600/150 as executed unless your own manifest proves it.

Evidence: An actual pilot allocation table and a split audit, including limitations. Future fitting/validation counts must be clearly distinguished from data already processed.

Role 4 - Tasks and handoffs, continued

Task 3: Analyze input-target relationships

Background: Time-series inputs are not ordinary demographic columns. RMS, peak absolute strain, spectral centroid and band powers are interpretable summaries, but can depend on preprocessing. Amplitude carries distance information jointly with other source parameters; whitening affects its scale. Training summaries must be derived consistently across runs.

Your task: Starting from Role 3 outputs, compute and document a small set of waveform summaries yourself. Join targets by ID. Create a feature-target correlation heatmap and at least one scatter plot such as RMS versus distance or spectral centroid versus chirp mass. Use O3a development data for choosing summaries and label any descriptive O4a inspection as exploratory. Discuss associations without causal or predictive claims.

Evidence: At least two detailed, labeled input-target plots with interpretations, plus summary definitions, units and exact N.

Task 4: Separate association from importance and summarize EDA

Background: A correlation measures association; permutation importance measures a fitted model's dependence on a feature under a specified evaluation procedure. The rubric asks for feature importance, so a correlation-only analysis leaves a real limitation. No training is required in this narrower checkpoint plan, and an incomplete requirement must not be disguised.

Your task: Combine raw-noise findings from Role 1, label/SNR selection findings from Role 2 and processing findings from Role 3. Explain skewness, outliers, parameter correlations and any examined time structure. Document whether model-based importance exists for this exact experiment; if not, mark it pending and specify a small validation-set permutation-importance study after baseline fitting. Do not import historical coefficients and call them new pilot importance.

Evidence: An EDA summary distinguishing completed descriptive evidence, untested hypotheses and the remaining feature-importance rubric gap.

Role 4 - Tasks and handoffs, continued

Task 5: Write the three-algorithm plan without fitting

Background: Linear regression is the main point-estimate baseline. Ridge adds coefficient regularization to this same direction. SNPE/MAF learns a joint conditional posterior. The assignment requires three algorithms, but preliminary model results are optional. The simulation prior and SNR selection must be understood before making formal SBC claims.

Your task: Describe each algorithm, why it fits the four-target problem and its intended improvement. Plan a small Ridge alpha grid selected on O3a validation, training-only scaling, a mean reference, per-target MAE/RMSE/R2 and documented normalized error. For SNPE, plan explicit validation settings, coverage/interval-width/failure reporting, and audit derived-target prior versus SNR-selected draws with Role 2. Explain that a past O4a inspection prevents calling that same dataset untouched. Keep all of this as a plan; do not train models in this notebook.

Evidence: A concise methodology document covering baseline plus two improvements, tuning, validation, comparison criteria and calibration assumptions, with no invented results.

Task 6: Assemble the revised milestone evidence

Background: The three-page PDF must include names/emails and actual individual contributions, alongside data, preprocessing, EDA and methodology. A notebook task scaffold is not a completed contribution. Existing historical artifacts can support claims only when their provenance is explicit.

Your task: Collect each owner's measured counts, short findings, labeled figures and contribution log. Choose a compact figure set with captions. Check every rubric item, including explicit non-applicable preprocessing items, exact dataset allocations, feature-importance status and data challenges. Leave names unassigned in this work plan; request actual ownership when the team chooses. Ensure one author per submitted notebook, numbering from 1 and the GitHub link. Keep old results and the manual pilot clearly separated. Do not include model scores or calibration outcomes in this milestone scope.

Evidence: A complete evidence checklist and short draft sections ready for the separate three-page PDF; unresolved requirements are flagged for team action.

Your exported files and acceptance contract

Outputs: analysis/dataset_audit.csv; analysis/split_counts.csv; analysis/feature_summary.csv; analysis/figures/; analysis/eda_notes.md; analysis/methodology_plan.md; analysis/milestone_checklist.md. No model checkpoints or fabricated scores.

File contract: dataset_audit.csv records sample_id, noise_id, group_id, run, split, injection_status, processing_status, rejection_reason and processed_row_index when present. split_counts.csv records experiment_id, run, split, n_source_groups, raw_seconds, raw_samples, attempted, accepted, rejected_snr, failed_waveform, rejected_processing and final_examples. feature_summary.csv records sample_id plus explicitly named numeric summaries with documented units.

Use the experiment-specific root artifacts/manual_preprocessing_milestone/<experiment_id>/ for all paths. Confirm sample IDs, source groups and retained counts with the next owner.

Combine the evidence and stop at the checkpoint

Milestone completion and handoff

Before marking your role complete, show the teammate consuming your artifacts one real record and its provenance. Give them a settings file, retained/rejected counts and a short interpretation, not just an array. Check sample_id, noise_id, group_id, run, split and shape at the handoff. Save outputs under your role-owned folder for the same experiment_id.

Deliver your completed notebook, a contribution log, 100-150 words of findings, figures with descriptive titles/axes/units/legends/captions and an explicit list of unfinished checks. Do not copy another person's implementation into your notebook. Preserve the older notebooks and existing PDFs.

Stop at this checkpoint: dataset preparation, validated preprocessing and EDA, exact allocation tables and the three-algorithm methodology plan. Full linear/Ridge/SNPE training, final model ranking, posterior calibration results and 800-per-run manual regeneration are not mandatory tasks in this set. Feature importance remains an acknowledged rubric gap if deferred; the team must decide how to complete that requirement before submission rather than asserting full compliance.

Write a final reflection: What did you personally implement? What evidence supports it? Which decisions need another owner's agreement? Which findings belong to the new pilot rather than the historical project? What remains before submission?

Three-page submission checklist

Include all full names and Berkeley emails; the motivation; actual raw/processed sizes and split counts; applied preprocessing and non-applicable items; distributions, input-target relationships, labeled plots and interpretations; time patterns if examined; data challenges; linear regression plus Ridge and SNPE plans; tuning/validation/metrics; actual individual contributions, notebook ownership and GitHub. Contributions count toward the three-page limit.

If exact counts, ownership or feature importance remain unfinished, flag them for completion before submission. Planned importance is an acknowledged rubric gap, not evidence of full compliance. Do not substitute old model scores for new manual-pilot findings.

Background reading

GWOSC public data

PyCBC waveforms

SciPy signal processing

sbi calibration guide

This teamwork guide is separate from the three-page milestone PDF. It contains background and tasks, not implementation solutions.